Questions by topic: Renewable energy technologies - Set 2 (15 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

0 1 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer ☐ times

0 1 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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0 1 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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0 1 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

0 1 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Extra space

15

END OF QUESTIONS